12TH BIOLOGY · CHAPTER 23 · FULL CHAPTER MCQS

Pharmacology

38 topic-wise MCQs from the complete chapter with answers and reasons

23.1 DRUG DISCOVERY AND DEVELOPMENT

1. The study of drugs and their effects on living organisms is:

(a) Pathology (b) Pharmacology
(c) Physiology (d) Immunology

Answer: (b) Pharmacology — It is a branch of biology.

2. Treatment with medicines prepared from plants is called:

(a) Chemotherapy (b) Gene therapy
(c) Radiotherapy (d) Phytotherapy

Answer: (d) Phytotherapy — Also called herbal medicine.

3. Confirming that the target plays an important role in the disease is:

(a) Target validation (b) Target identification
(c) Lead optimization (d) Clinical trial

Answer: (a) Target validation — It is the second stage of drug discovery.

4. Promising substances that interact effectively with the target are:

(a) Antibodies (b) Placebos
(c) Lead compounds (d) Antigens

Answer: (c) Lead compounds — Found by screening thousands of compounds.

5. Modifying lead compounds to reduce toxicity is called:

(a) Target validation (b) Lead optimization
(c) Surveillance (d) Marketing

Answer: (b) Lead optimization — It also improves effectiveness.

6. Testing of compounds in cell-based culture is called:

(a) In vivo (b) Phase 3
(c) Phase 1 (d) In vitro

Answer: (d) In vitro — In vivo testing is done in animals.

7. Phase 1 clinical trials are carried out on:

(a) 20–100 healthy volunteers (b) 1,000–3,000 patients
(c) 100–300 patients (d) Animals

Answer: (a) 20–100 healthy volunteers — To determine safety and dosage.

studysapiens.com

8. Phase 3 trials involve about:

(a) 20–100 people (b) 100–300 people
(c) 1,000–3,000 people (d) 10 people

Answer: (c) 1,000–3,000 people — To confirm efficacy and monitor adverse reactions.

9. Which agency reviews drugs before they reach the market?

(a) WHO only (b) NASA
(c) UNESCO (d) FDA or EMA

Answer: (d) FDA or EMA — They approve or reject the drug.

10. Monitoring a drug after its release is called:

(a) Post-marketing surveillance (b) Preclinical research
(c) Lead optimization (d) Target validation

Answer: (a) Post-marketing surveillance — For long-term and rare side effects.

23.2 CLASSES OF ANTIBIOTIC

11. Antibiotics are used against infections caused by:

(a) Viruses (b) Bacteria
(c) Prions (d) Allergens

Answer: (b) Bacteria — They kill bacteria or stop their growth.

12. Penicillin was discovered by:

(a) Gerhard Domagk (b) Kary Mullis
(c) George Lesher (d) Alexander Fleming

Answer: (d) Alexander Fleming — In 1928.

13. Penicillin is derived from:

(a) Streptomyces (b) Cinchona bark
(c) Penicillium fungi (d) Sulfanilamide

Answer: (c) Penicillium fungi — It is a bactericidal antibiotic.

14. Penicillin acts by inhibiting bacterial:

(a) Cell wall synthesis (b) Protein synthesis
(c) DNA replication (d) Folate synthesis

Answer: (a) Cell wall synthesis — It stops cross-linking of peptidoglycan.

15. The cell wall chains whose cross-linking is stopped by penicillin are made of:

(a) Cellulose (b) Keratin
(c) Chitin (d) Peptidoglycan (murein)

Answer: (d) Peptidoglycan (murein) — The cell then bursts.

16. Bacteria resist penicillin by producing:

(a) Beta-lactamases (b) Lysozyme
(c) Topoisomerases (d) Folate

Answer: (a) Beta-lactamases — These are specific enzymes.

17. Tetracycline binds to which ribosomal subunit of bacteria?

(a) 50S (b) 30S
(c) 60S (d) 80S

Answer: (b) 30S — The binding is reversible.

18. Tetracycline should not be taken with:

(a) Water (b) Fruit
(c) Dairy or iron supplements (d) Rice

Answer: (c) Dairy or iron supplements — As advised in the textbook.

19. Tetracycline is used to treat infections such as:

(a) Chlamydia and syphilis (b) Influenza
(c) Malaria (d) HIV

Answer: (a) Chlamydia and syphilis — They are bacterial infections.

20. An antibiotic that stops growth without killing bacteria is:

(a) Bactericidal (b) Bacteriostatic
(c) Antiviral (d) Antiseptic

Answer: (b) Bacteriostatic — For example tetracycline.

21. Fluoroquinolones were discovered by:

(a) Alexander Fleming (b) Gerhard Domagk
(c) George Lesher (d) Robert Koch

Answer: (c) George Lesher — In 1962.

22. Fluoroquinolones inhibit bacterial:

(a) Cell wall synthesis (b) Folate synthesis
(c) Protein synthesis (d) DNA replication

Answer: (d) DNA replication — They inhibit type II topoisomerases.

23. In Gram-negative bacteria, fluoroquinolones mainly target:

(a) DNA gyrase (b) Topoisomerase IV
(c) 30S ribosome (d) Peptidoglycan

Answer: (a) DNA gyrase — DNA uncoiling stops.

24. Quinine is a natural alkaloid from the bark of:

(a) Neem tree (b) Cinchona tree
(c) Willow tree (d) Pine tree

Answer: (b) Cinchona tree — A South American tree.

studysapiens.com

25. Tendon rupture is a rare side effect of:

(a) Penicillin (b) Tetracycline
(c) Fluoroquinolones (d) Sulfonamides

Answer: (c) Fluoroquinolones — Also peripheral neuropathy and cardiac issues.

26. Sulfonamides were discovered in 1932 by:

(a) Fleming (b) Pasteur
(c) Lesher (d) Gerhard Domagk

Answer: (d) Gerhard Domagk — They are derived from sulfanilamide.

27. Sulfonamides inhibit the bacterial synthesis of:

(a) Cell wall (b) Folic acid
(c) Proteins (d) Lipids

Answer: (b) Folic acid — Folate is essential for DNA synthesis.

28. Sulfonamides are often combined with:

(a) Penicillin (b) Quinine
(c) Trimethoprim (d) Insulin

Answer: (c) Trimethoprim — For a stronger bactericidal effect.

29. Sulfonamides are primarily used for:

(a) Urinary tract and skin infections (b) Viral fever
(c) Diabetes (d) Cancer

Answer: (a) Urinary tract and skin infections — Also acne and respiratory infections.

30. Kidney damage is a possible side effect of:

(a) Penicillin (b) Monoclonal antibodies
(c) Tetracycline (d) Sulfonamides

Answer: (d) Sulfonamides — Also rashes, nausea and photosensitivity.

23.3 – 23.4 ANTIVIRALS AND MONOCLONAL ANTIBODIES

31. Drugs used to treat viral infections are called:

(a) Antivirals (b) Antibiotics
(c) Antifungals (d) Antipyretics

Answer: (a) Antivirals — They prevent the multiplication of viruses.

32. Antiretroviral drugs are designed to treat infection by:

(a) Influenza virus (b) Herpes virus
(c) HIV (d) Hepatitis virus

Answer: (c) HIV — HIV is a retrovirus.

studysapiens.com

33. Retroviruses replicate by using the enzyme:

(a) DNA polymerase (b) Lysozyme
(c) Ligase (d) Reverse transcriptase

Answer: (d) Reverse transcriptase — Their genetic material is RNA.

34. Which statement is correct?

(a) All antivirals are antiretrovirals (b) All antiretrovirals are antivirals
(c) Antivirals kill bacteria (d) Antiretrovirals treat influenza

Answer: (b) All antiretrovirals are antivirals — Retroviruses are one group of viruses.

35. Monoclonal antibodies are:

(a) Natural plant extracts (b) Vitamins
(c) Synthetic antibiotics (d) Laboratory-produced molecules

Answer: (d) Laboratory-produced molecules — They enhance the immune system's ability.

36. Monoclonal antibodies bind to:

(a) Any antigen (b) Bacterial cell walls only
(c) A single specific target (d) No target

Answer: (c) A single specific target — They have high specificity.

37. In cancer, monoclonal antibodies provide:

(a) Targeted therapy (b) Broad-spectrum therapy
(c) Bacteriostatic therapy (d) Herbal therapy

Answer: (a) Targeted therapy — They reduce damage to healthy tissues.

38. Compared with traditional drugs, monoclonal antibodies have:

(a) Higher toxicity (b) Lower toxicity
(c) No effect (d) Shorter effects

Answer: (b) Lower toxicity — With potential for long-lasting effects.

studysapiens.com